



CI/CD Pipeline using Jenkins – Practical Record

AIM

To implement a CI/CD pipeline using Jenkins for a Java web application and deploy it to Apache Tomcat with artifact storage in Artifactory.

REQUIREMENTS

- JDK 17
 - Apache Maven
 - Jenkins
 - Apache Tomcat 9
 - JFrog Artifactory
 - SonarQube
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PROCEDURE

Step 1: Create Maven Project

1. Create a Maven project.
 2. Set packaging type as `war`.
 3. Configure `pom.xml` with servlet dependency.
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Step 2: Develop Application

1. Create a Servlet (`App.java`) to perform calculations.
 2. Create `index.jsp` for user input.
 3. Map servlet using `@WebServlet`.
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Step 3: Build the Project

Execute:

```
mvn clean package
```

Output: target/calculator.war

Step 4: Configure Tomcat

1. Copy WAR file to webapps/.
2. Start Tomcat using:

startup.bat

3. Access:

http://localhost:8080/calculator

Step 5: Enable Tomcat Manager

1. Edit tomcat-users.xml
2. Add roles:

```
<role rolename="manager-gui"/>
<role rolename="manager-script"/>
```

3. Add user:

```
<user username="tomcat" password="s3cret"
roles="manager-gui,manager-script"/>
```

Step 6: Configure Artifactory

1. Create repositories:
 - libs-release-local
 - libs-snapshot-local
2. Add in pom.xml:

```
<distributionManagement>
  <repository>
    <id>artifactory</id>
    <url>http://localhost:8081/artifactory/libs-
release-local</url>
  </repository>
</distributionManagement>
```

3. Configure settings.xml:

```
<servers>
  <server>
    <id>artifactory</id>
    <username>admin</username>
    <password>password</password>
  </server>
</servers>
```

Step 7: Create Jenkins Jobs

Create a folder: **Central_CI**

Job 1: Setup

- Archive artifacts: **/*
 - Trigger next job: Compile
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Job 2: Compile

- Copy artifacts from Setup
- Maven goal:

compile

- Trigger: Unit Test
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Job 3: Unit Test

- Copy artifacts from Compile
- Maven goal:

test

Job 4: Code Coverage

- Copy artifacts
- Maven goal:

`jacoco:report`

- Trigger: Static Analysis
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Job 5: Static Analysis

- Copy artifacts
- Maven goal:

`sonar:sonar -Dsonar.token=TOKEN`

- Trigger: WAR
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Job 6: WAR

- Copy artifacts
- Maven goal:

`war:war`

- Trigger: ToArtifactory
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Job 7: ToArtifactory

- Copy artifacts
- Maven goal:

`deploy`

- Trigger: Smoke Test
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Job 8: Smoke Test

1. Copy WAR file:

`**/*.war`

2. Deploy using Jenkins:

- Tomcat URL: `http://localhost:8080`
- Credentials:
Username: tomcat
Password: s3cret

Step 8: Create Pipeline View

1. Install **Build Pipeline Plugin**
2. Create new view → Build Pipeline View
3. Select initial job: Setup

FLOW DIAGRAM

Setup → Compile → Unit Test → Code Coverage →
Static Analysis → WAR → Artifactory → Smoke Test

RESULT

- WAR file created successfully
- Application deployed in Tomcat
- Artifacts stored in Artifactory
- Pipeline executed successfully

CONCLUSION

Thus, a complete CI/CD pipeline was implemented using Jenkins, Maven, SonarQube, Artifactory, and Tomcat.

OUTPUT

`http://localhost:8080/calculator`
